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HAIQU COGNITIVE TEST DESIGN: FOR DISCUSSION

DECEMBER 16, 2025 | PRIVATE & CONFIDENTIAL

RECOMMENDED PLATFORM: TESTABLE

- Device capabilities
  - Desktop computer
  - Tablets
- Features
  - Test design and implementation
  - Results export
- Library
  - Extensive library from Testable and other researchers
  - Can also build own
- Pricing
  - Either $299/year for Plus, $699/year for Pro for remote multi-site, multi-participant tests
  - Implementation: $300-400/day, with clear specs
  - Consultation: 1-2 days for study and experiment design

LITERATURE BACKGROUND: RESPIRATORY ILLNESS AND COGNITION

- Acute respiratory infections, even without fever, are associated with measurable changes in cognitive performance, particularly processing speed, attention, and memory retrieval
- Effects are often subtle, best detected using within-person repeated measures
- Practice effects are common in cognitive testing, requiring prospective baseline designs

Key findings from prior studies

- Slower reaction times during illness
- Effects observed in working adults, not just clinical populations
- Time-of-workday testing matters

Implications for HAIQU

- Cognitive impacts are plausible, measurable, and relevant to healthcare staff
- Repeated testing must be fast and operationally feasible
- Lack of large body of literature suggests potentially innovative research, especially in clinical settings

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INCLUDE COGNITIVE PERFORMANCE FOR ANOTHER DIMENSION OF ROI ANALYSIS

Design principle:

Use validated, objective cognitive tasks rather than developing new questionnaires or instruments.

Respiratory illness may affect attention, vigilance, and decision speed, all critical in clinical environments

Objective cognitive measures provide

Complementary data to symptom and exposure tracking

A functional outcome that can be linked to environmental and operational variables

Aligns with epidemiologic interest in transient effects, not just severe illness

Design constraints

Must be short (≤ 10 minutes)

Computer-based

Multi-site scale

Suitable for repeated administration

Scientifically credible and defensible

COGNITIVE TEST OPTIONS

Option 1: NIH Toolbox Cognitive Tests

- NIH-developed, standardized cognitive battery
- Designed for epidemiologic studies and clinical trials (ages 3–85

Relevant tests

- Pattern Comparison → processing speed
- Flanker Task → inhibitory control and selective attention
- Dimensional Change Card Sort (DCCS) → cognitive flexibility

Strengths

- Strong validation in adults
- Widely used and recognized in NIH-funded research
- Designed for repeated measures and population studies
- Minimal need to justify construct validity

Option 2: Modular Research-Standard Battery

Assemble a battery from well-established cognitive paradigms commonly used in occupational and applied cognition research

Example test construct

- Psychomotor Vigilance Test (PVT, 3 min): Sustained attention / alertness
- Processing speed task (2–3 min): Simple choice reaction time
- Go / No-Go task (3 min): Inhibitory control and selective attention

Strengths

- Covers vigilance, processing speed, and executive control
- Suitable for frequent repeated measures
- Low or no licensing cost
- Flexible integration with HAIQU workflows and AI-guided timing

PLATFORM OPTIONS & COST COMPARISON

	Platform / Option	Cognitive Domains Covered	Typical Tests	Administration	Time per Session	Cost Structure	Key Strengths	Key Limitations	Link
Option 1	NIH Toolbox	Processing speed, executive function	Pattern Comparison, Flanker, DCCS	iPad app (licensed)	6–10 min	$600–$2,500 per year, per device tier (1–2, 3–6, 7–10 devices)	Strong validation, widely accepted in epi and clinical research	Cost scales with number of devices and locations, app-based workflow, limited customization	https://nihtoolbox.org/get-the-toolbox/
Option 2	Hosted Research Platform (Testable)	Vigilance, processing speed, executive control	RT tasks, Go/No-Go	Browser (hosted)	8–9 min	~$700 per year (tier-dependent)	Minimal setup, scalable, browser-based, peer-reviewed	Ongoing subscription, platform dependency	https://www.testable.org/#how_it_works
	PsyToolkit Experiment Library	Processing speed, inhibition, attention	Go/No-Go, RT, Flanker, Stroop, others	Browser-based	≤10 min	$0 (academic use)	Ready-to-use tasks, well-documented, free for research	Less customizable UI, hosted externally	https://www.psytoolkit.org/experiment-library
	Open-Source Modular Battery (Custom Assembly)	Vigilance, processing speed, inhibitory control	PVT, Choice RT, Go/No-Go, and others common in cognitive	Browser-based or local	8–9 min	$0	Free, open source, flexible, well-established paradigms, scalable across sites	Requires setup and documentation	https://github.com/flowersteam/cognitive-testbattery
	PEBL (Psychology Experiment Building Language)	Broad cognitive coverage	Classic paradigms (RT, inhibition, memory)	Local desktop install only	Variable (3–10 min)	$0	Open source, comprehensive, long research history	Local install only, less flexible UI, not browser-native	https://pebl.sourceforge.net/

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SELECTED LITERATURE SUMMARY

Study	Study Design / Population	Key Cognitive Findings	Cognitive Domains Affected	Tests / Platforms Used	Relevance to HAIQU
Bucks et al., 2008 "Selective effects of upper respiratory tract infection on cognition, mood and emotion processing: A prospective study"	Prospective baseline + retest during natural URTI; younger and older adults (N=80, control vs exp)	Ill participants showed attenuated practice effects. Greater symptom severity linked to slower memory retrieval.	Processing speed, attention, memory retrieval	Cognitive Drug Research (CDR) computerized battery	Supports within-person repeated testing and short computerized tasks
Smith, 2023 "Effects of Mild URTI on Mood and Cognition"	Repeated measures; adults with mild URTI vs healthy controls (N=60)	Reduced alertness; slower simple and choice RT; impaired sustained attention	Vigilance, processing speed, attention	Simple RT, Choice RT, sustained attention task	Supports short RT and vigilance tasks for mild respiratory illness
Smith et al., 2000 "Effects of upper respiratory tract illnesses on mood and performance over the working day"	Experimental and observational adult studies	Slower simple and choice RT; reduced alertness; effects worsened during work day	Processing speed, vigilance	Simple RT, Choice RT, mood rating	Highlights work shift timing-sensitive of tests during illness
Smith, 2012 Review: "Twenty-five years of research on the behavioural malaise associated with influenza and the common cold"	Review of 25+ years of adult studies	Illness-related malaise degrades performance and increases sensitivity to other stressors	Vigilance, attention, cognitive resilience	Review	Highlights compounded risk during illness in safety-critical work
Zelazo et al., 2014 "NIH Toolbox Cognition Battery (CB): Validation of Executive Function Measures in Adults"	Adult validation study across ages 20–85 (N=268), multi-site	Executive function measures showed strong reliability, validity, and sensitivity to age	Inhibitory control, selective attention, cognitive flexibility	NIH Toolbox: Flanker, DCCS	Establishes NIH Toolbox EF tasks as credible for epidemiologic studies

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Smith, 2012 "Behavioral Effects of Upper Respiratory Tract Illnesses: A Consideration of Possible Underlying Cognitive Mechanisms"	Narrative review (adults)	Cognitive effects observed even in mild illness, distinct from mood alone	Attention, processing speed	RT and attention tasks	Reinforces sensitivity of short performance tests

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